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Experiment 4:

Build an Artificial Neural Network by implementing the Backpropagation algorithm and test the same using appropriate data sets.

Code:

```
import numpy as np
X = np.array([[0,0],[0,1],[1,0],[1,1]])
y = np.array([[0],[1],[1],[0]])
sig = lambda x: 1/(1+np.exp(-x))
dsig = lambda x: x*(1-x)
np.random.seed(1)
w1 = np.random.rand(2,2)
w2 = np.random.rand(2,1)
for i in range(10000):
    h = sig(X @ w1)
    o = sig(h @ w2)
    d2 = (y - o) * dsig(o)
    d1 = (d2 @ w2.T) * dsig(h)
    w2 += h.T @ d2 * 0.5
    w1 += X.T @ d1 * 0.5
print("Predicted Output:")
print(np.round(o,4))
print("\nClassified Output:")
print((o>0.5).astype(int))
```

Output:

```
Predicted Output:
[[0.0565]
 [illegible]52]
 [0.8952]
 [0.1392]]

Classified Output:
[[0]
 [1]
 [1]
 [0]]
```